

Mongosh Practical(DBMS)

ASHWIK(2520030224)

```
Using MongoDB:      8.3.8
Using Mongosh:      2.10.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-09-02T11:57:35.375+05:30: Access control is not enabled
for the database. Read and write access to data and configuratio
n is unrestricted
-----

test> use BookFlow
switched to db BookFlow
BookFlow>
```

1st: Document Modelling

```
BookFlow> db.book_metadata.insertOne([
  {
    title: "The Silent Algorithm",
    author: "R. Menon",
    genre: "Technology",
    publishedYear: 2022,
    formats: ["Hardcover", "Digital", "Audiobook"],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: "K. Rao",
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: "M1001",
        rating: 5,
        comment: "Excellent deep dive into digital systems design.",
        date: ISODate("2023-01-15")
      },
      {
        member_id: "M1042",
        rating: 4,
        comment: "Good but the Digital edition had formatting issues.",
        date: ISODate("2023-02-20")
      }
    ]
  }
]);

{
  acknowledged: true,
  insertedId: ObjectId('6a97f41a37c2df90481cd840')
}
```

```
bookFlow> db.book_metadata.insertMany([
  {
    title: "Ocean of Code",
    author: "A. Fernandes",
    genre: "Technology",
    publishedYear: 2021,
    formats: ["Digital", "Paperback"],
    specs: {
      pages: 210,
      fileSize_MB: 2.1,
      drm: false
    },
    avgRating: 4.2,
    reviews: [
      {
        member_id: "M2001",
        rating: 4,
        comment: "Solid Digital read, clean layout.",
        date: ISODate("2022-05-10")
      },
      {
        member_id: "M2002",
        rating: 5,
        comment: "Best programming book this year.",
        date: ISODate("2022-06-01")
      }
    ]
  },
  {
    title: "Whispers of Yesterday",
    author: "S. Kapoor",
    genre: "Fiction",
    publishedYear: 2019,
    formats: ["Hardcover", "Paperback"],
    specs: {
      pages: 310

```

```
    binding: "Sewn"
  },
  avgRating: 3.8,
  reviews: [
    {
      member_id: "M3001",
      rating: 3,
      comment: "Slow start but strong ending.",
      date: ISODate("2020-01-15")
    },
    {
      member_id: "M3002",
      rating: 4,
      comment: "Beautifully written prose.",
      date: ISODate("2020-02-11")
    }
  ]
},
{
  title: "Quantum Horizons",
  author: "R. Menon",
  genre: "Science",
  publishedYear: 2023,
  formats: ["Digital", "Audiobook"],
  specs: {
    fileSize_MB: 5.6,
    drm: true,
    narrator: "P. Iyer",
    runtime_min: 540
  },
  avgRating: 4.9,
  reviews: [
    {
      member_id: "M4001",
      rating: 5,
      comment: "The Digital format made annotations so easy."

```

```

      date: ISODate("2023-08-01")
    },
    {
      member_id: "M4002",
      rating: 5,
      comment: "Mind-bending and well narrated.",
      date: ISODate("2023-09-14")
    }
  ]
}
});

acknowledged: true,
insertedIds: {
  '0': ObjectId('6a97f44b37c2df98481cd841'),
  '1': ObjectId('6a97f44b37c2df98481cd842'),
  '2': ObjectId('6a97f44b37c2df98481cd843')
}
}

```

```

BookFlow> db.book_metadata.find().pretty()
[
  {
    _id: ObjectId('6a97e0404f6580400309a236'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 300,
      fileSize_MB: 4.2,
      drw: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.'
      },
      {
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a97f41a37c2df98481cd840'),
    title: 'The Silent Algorithm',
    author: 'R. Menon'
  }
]

```

```

    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 0.2,
      drw: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 0.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.'
      },
      {
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a97f44b37c2df98481cd841'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2023,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drw: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.'
      }
    ]
  }
]

```

```

    },
    {
      member_id: 'M0002',
      rating: 4,
      comment: 'Beautifully written prose.',
      date: ISODate('2020-02-11T00:00:00.000Z')
    }
  ],
  {
    _id: ObjectId('6a97f40b37c2d0f90481cd8013'),
    title: 'Quantum Horizons',
    author: 'R. Penon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drw: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M0001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M0002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
]

```

2nd Advanced Querying

```

BookFlow> db.book_metadata.find(
|   { avgRating: { $gt: 4 } },
|   { title: 1, avgRating: 1, _id: 0 }
| );
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },
  { title: 'Quantum Horizons', avgRating: 4.9 }
]
BookFlow>

```

```

BookFlow> db.book_metadata.find(
|   { avgRating: { $gt: 4 } },
|   { title: 1, avgRating: 1, _id: 0 }
| );
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },
  { title: 'Quantum Horizons', avgRating: 4.9 }
]
BookFlow> db.book_metadata.find(
|   { formats: { $in: [ "Digital", "Audiobook" ] } },
|   { title: 1, formats: 1, _id: 0 }
| );
[
  {
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  { title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
  { title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }
]
BookFlow>

```

```
BookFlow> db.book_metadata.find([
|   avgRating: { $gt: 4 },
|   formats: { $in: ["Digital"] }
| ]);
[
  {
    _id: ObjectId('6a97e0404fb580440309a236'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 300,
      fileSize_MB: 4.2,
      drw: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ],
  },
  {
    _id: ObjectId('6a97f41a37c2df90481cd840'),
```

```
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 300,
      fileSize_MB: 4.2,
      drw: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ],
  },
  {
    _id: ObjectId('6a97f44b37c2df90481cd841'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drw: false },
    avgRating: 4.2,
    reviews: [
```

```

    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 300,
      fileSize_MB: 4.2,
      drw: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ],
  },
  {
    _id: ObjectId('6a97f44b37c2df90481cd841'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drw: false },
    avgRating: 4.2,
    reviews: [

```

```

BookFlow> db.book_metadata.find(
|   { formats: { $regex: "Digital", $options: "i" } },
|   { title: 1, formats: 1 }
| );
[
  {
    _id: ObjectId('6a97e0404fb580440309a236'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    _id: ObjectId('6a97f41a37c2df90481cd840'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    _id: ObjectId('6a97f44b37c2df90481cd841'),
    title: 'Ocean of Code',
    formats: [ 'Digital', 'Paperback' ]
  },
  {
    _id: ObjectId('6a97f44b37c2df90481cd843'),
    title: 'Quantum Horizons',
    formats: [ 'Digital', 'Audiobook' ]
  }
]
BookFlow> |

```

```

BookFlow> db.book_metadata.find(
| { "reviews.comment": { $regex: "Digital", $options: "i" } },
| { title: 1, "reviews.$": 1 }
| );
{
  {
    _id: ObjectId('6a97e0404fb580448309a236'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a97f41a37c2df90481cd840'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a97f44b37c2df90481cd841'),
    title: 'Ocean of Code',
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',

```



3rd TASK: Aggregation Pipeline



4th TASK: Indexing for Speed













